Food Landscape of Ho Chi Minh City

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1 Introduction

This report presents a data-driven analysis of the food landscape across 19 districts of Ho Chi Minh City (HCMC). Using OpenStreetMap (OSM) data, we collected 9,707 food places including restaurants, cafes, and fast food outlets. The analysis explores spatial distribution patterns, cuisine diversity, and digital representation inequality across districts.

2 Data Collection

Data was collected via the Overpass API (OpenStreetMap) covering 19 districts with adaptive radius sampling — smaller radii for dense central districts and larger radii for sprawling outer districts. The final dataset contains 9,707 food places with attributes including district, name, cuisine type, amenity type, and geographic coordinates.

Metric	**Value**
Total places	9,707
Districts covered	19
Restaurant	5,377 (55.4%)
Cafe	3,585 (36.9%)
Fast food	745 (7.7%)

Table 1: Dataset Summary

Data Representativeness

It is important to note that OSM data only captures food places that have been *manually mapped by contributors* — typically business owners, tech-savvy customers, or food bloggers. This introduces a systematic bias:

- Places in central, tourist-facing districts (Quan 1, Quan 3) are more likely to be mapped than local eateries in working-class areas.
- Street vendors, unlicensed stalls, and informal food spots are almost entirely absent from the dataset.
- The dataset therefore reflects **digital food presence**, not the actual number of food establishments.

This limitation is acknowledged throughout the analysis and is itself treated as a finding revealing the *digital divide* in how Saigon's food scene is represented online.

3 Exploratory Data Analysis

3.1 Distribution by District

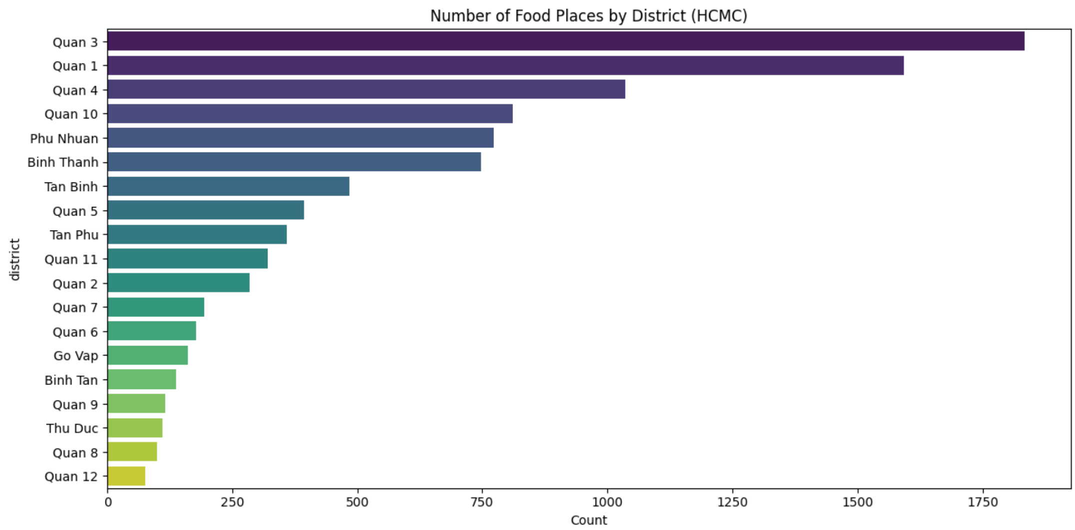

Figure 1: Number of food places by district. Quan 3 and Quan 1 dominate with over 1,800 and 1,500 places respectively.

Central districts (Quan 1, Quan 3, Quan 4) show significantly higher food place density, reflecting both higher actual density and stronger digital presence among their customer base. Quan 1 and Quan 3 in particular benefit from active food blogger communities, tourist foot traffic, and business owners who actively maintain their OSM listings.

Quan 4, despite being a working-class district with narrow alleyways, shows surprisingly high counts — suggesting that proximity to the city center drives digital mapping behavior even in non-touristy areas.

In contrast, several districts show unexpectedly low counts despite high actual food density:

- **Quan 5** — home to Cho Lon's long-established Chinese-Vietnamese food culture, where many family-run eateries have operated for decades without a digital presence.
- **Thu Duc** — despite hosting the largest university cluster in Vietnam, much of its student food scene physically lies across the Binh Duong provincial border (DHQG campus area), outside our sampling scope.
- **Quan 9** — predominantly an industrial zone where factory workers dine in company canteens rather than mapped establishments.
- **Binh Tan** — while the district hosts major commercial centers (Aeon Mall, Duong Ten Lua corridor), many food establishments are tagged under `food_court` or `mall` in OSM rather than `restaurant` or `cafe`, causing systematic undercounting. This tag mismatch affects all districts to varying degrees but is most pronounced here.

Notably, Phu Nhuan captures 773 places with a sampling radius of only 2,500m, while Binh Thanh yields 747 places despite a larger radius of 3,500m. This suggests Phu Nhuan may have genuinely higher food place density, consistent with its reputation as one of Saigon's most vibrant cafe and dining hubs. However, normalized density metrics would be needed to confirm this observation conclusively.

3.2 Food Place Types

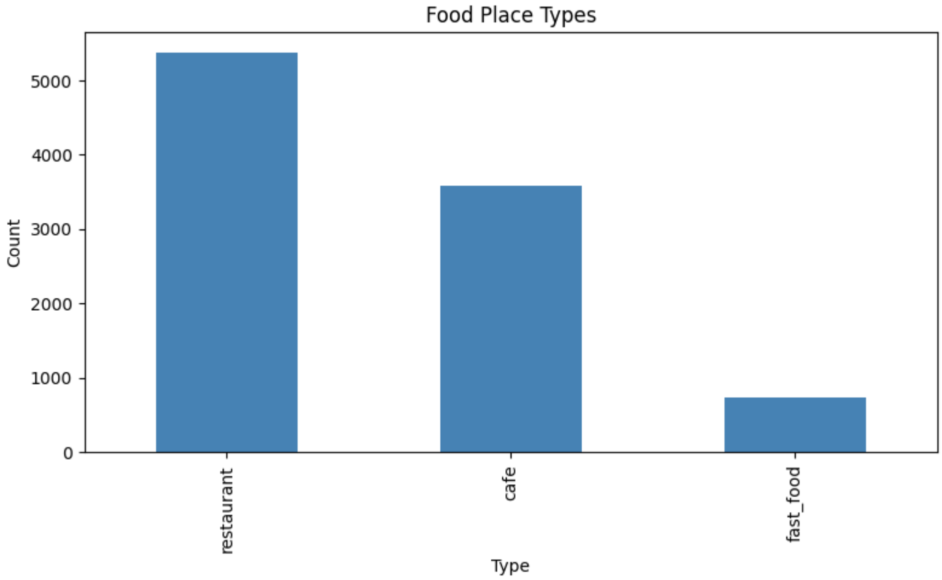

Figure 2: Distribution of food place types across HCMC.

Despite Saigon's reputation as a cafe city, restaurants outnumber cafes by a ratio of 1.5:1, suggesting that the cafe culture perception may be driven by social media visibility rather than actual density.

3.3 Cuisine Distribution by District

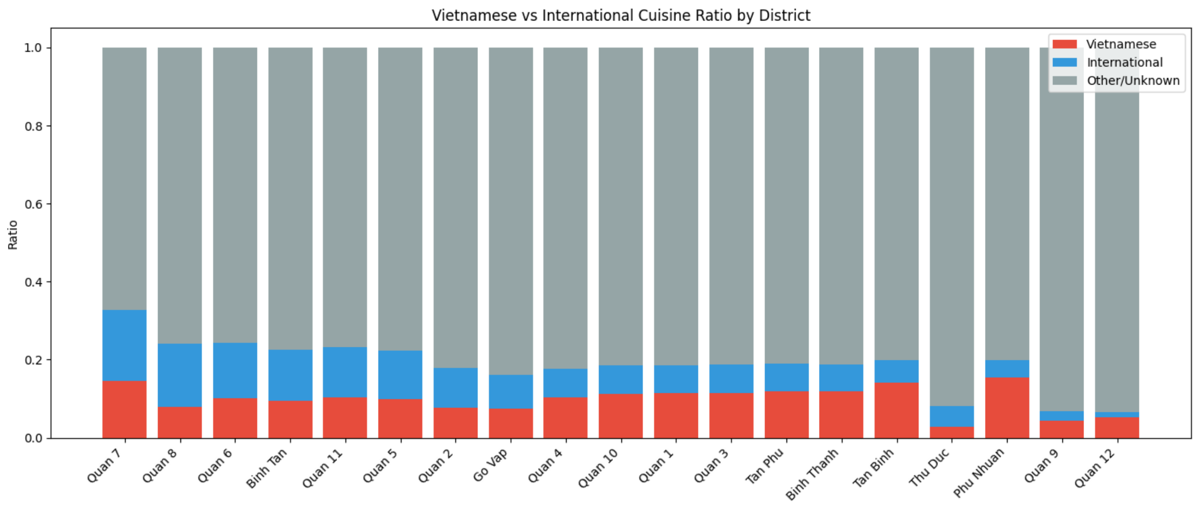

Figure 3: Vietnamese vs International cuisine ratio by district, sorted by international ratio.

The stacked bar chart reveals clear patterns in cuisine diversity across districts:

– **Quan 7** shows the highest international cuisine ratio, consistent with its Phu My Hung urban area which hosts a large expat community from Korea, Japan, and Taiwan — driving demand for international dining options.

– **Quan 2** also shows elevated international presence, reflecting the Thao Dien expat community with predominantly Western dining preferences.

– **Quan 6** international ratio is partly explained by its Chinese-Vietnamese community — Chinese cuisine is classified as international in this analysis, inflating the ratio relative to purely Vietnamese districts.

– **Thu Duc** shows a surprisingly high international ratio, potentially driven by international student communities at its university cluster.

– **Quan 9 and Quan 12** are dominated by the grey "Other/Unknown" category, confirming their status as the most underrepresented districts in terms of data quality.

Vietnamese cuisine remains dominant in traditional central districts such as Quan 3 and Quan 4, while international cuisine clusters around affluent and expat-heavy areas (Quan 7, Quan 2), reflecting the socioeconomic geography of the city.

4 Clustering Analysis

4.1 PyTorch Autoencoder Embeddings

District embeddings were learned using a PyTorch Autoencoder trained on 9 food features per district. The model compressed each district into a 4-dimensional embedding, which was then clustered using KMeans (k=4).

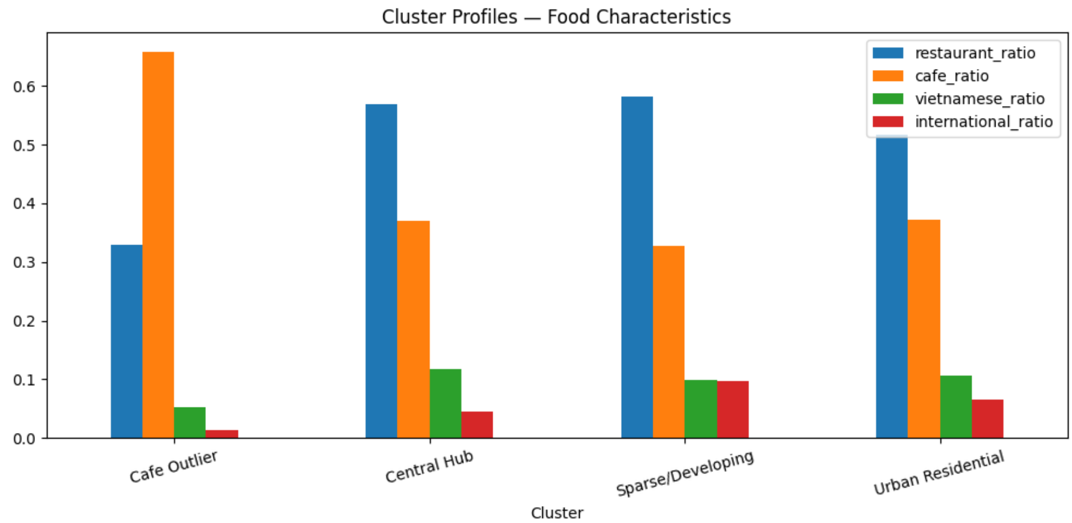

Figure 4: Cluster profiles showing food characteristics per cluster.

The Cafe Outlier cluster (Quan 12) stands out with a `cafe_ratio` of ∼0.66 — significantly higher than all other clusters. This anomaly is partly genuine, as Quan 12 appears to have a localized concentration of cafes, but may also be amplified by OSM tagging inconsistency: many Vietnamese cafes are tagged as either `coffee_shop` or `coffee_shop;vietnamese` interchangeably, depending on the contributor. Both tags describe the same type of establishment, but in our feature extraction they are treated as separate amenity signals, potentially inflating the perceived cafe dominance in districts where this dual-tagging is more common.

The Central Hub cluster (Quan 1, Quan 3, Quan 4, Binh Thanh, Phu Nhuan) shows the highest `vietnamese_ratio` ($\sim$0.11) among the four clusters, consistent with these districts being the traditional heartland of Vietnamese dining culture in the city.

The Sparse/Developing cluster shows the highest `international_ratio` ($\sim$0.10), driven primarily by Quan 7 (Phu My Hung) and Quan 2 (Thao Dien) — both known expat zones — pulling the cluster average up despite the other members being relatively underdeveloped in terms of food data coverage.

4.2 Cluster Descriptions

Cluster	**Districts**	**Characteristics**
Central Hub	Q1, Q3, Q4, Binh Thanh, Phu Nhuan	High density, diverse cuisine
Urban Residential	Q5, Q10, Q11, Tan Binh, Go Vap	Mixed, moderate density
Sparse/Developing	Q2, Q6, Q7, Q8, Binh Tan	Low density, high international
Cafe Outlier	Quan 12	Unusually high cafe ratio

Table 2: District Cluster Descriptions

5 Key Findings

- **Digital inequality exists** — OSM coverage strongly correlates with district socioeconomic profile, not actual food density.
- **Central districts dominate** — Quan 1 and Quan 3 account for over 35% of all mapped food places.
- **Quan 7 is a unique outlier** — Low quantity but highest data quality, driven by expat community behavior.
- **Quan 12 cafe anomaly** — Disproportionately high cafe ratio suggests a localized cafe cluster.
- **Restaurant > Cafe** — Despite perception, restaurants outnumber cafes 1.5:1 across the city.

6 Limitations

- OSM data reflects *digital presence*, not actual restaurant count.
- Radius-based sampling may cause overlap between adjacent districts.
- Future work: supplement with Foody/GrabFood data for ground truth comparison.